

Engineering Mathematics-II

Practice Set: Linear Algebra and Inner product space.

1. Show that the polynomials of degrees less than or equal to 3 with real coefficients form a vector space.
2. Discuss whether $\mathbb{R}^2(\mathbb{R})$ is a subspace of $\mathbb{R}^3(\mathbb{R})$.
3. Let $M_{(2 \times 2)}(\mathbb{R})$ be the set of real matrices of order 2×2 . Is W a subspace of $M_{(2 \times 2)}(\mathbb{R})$, where W is the set of matrices of the form $\begin{bmatrix} a & a+b \\ a & 10b \end{bmatrix}$.
4. Describe geometrically span (u), where u is a non-zero vector in $\mathbb{R}^3(\mathbb{R})$.
5. Determine whether the set of vectors $\{(2, -5, 3, 4), (1, -2, 3, 5), (2, 1, -1, 5)\}$ is linearly dependent or independent.
6. In $P_2(x)$, determine whether $r(x) = 1 - 4x + 6x^2$ is in the span of $(p(x), q(x))$, where $p(x) = 1 - x + x^2$ and $q(x) = 2 + x - 3x^2$.
7. Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 3 \\ x & y & z \end{bmatrix}$ and $S = \{(x, y, z) \in \mathbb{R}^3 : \det(A) = 0\}$. Is S forms a subspace of $M_{(3 \times 3)}(\mathbb{R})$? If yes, find a basis and dimension of S .
8. Let mapping $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be defined by $T(x, y) = (x + y, 2x + 3y, x - y)$. Is T a linear transformation?
9. Show that the map $F: \mathbb{R}^2 \rightarrow \mathbb{R}^3$, defined by $F(x, y) = (x, 2x - y, 3x + 4y)$ is a linear transformation. Find the kernel, nullity, and rank of F .
10. Let the linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be defined by $T(x, y, z) = (x + y - z, x + y + z, y - z)$. Find the matrix of linear transformation T with respect to the ordered basis $B = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$.
11. Let the linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ and $S: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be defined by $T(x, y) = (x, 2x - y, 3x + 4y)$ and $S(x, y, z) = (2x + z, 3x - z, x + y + z)$. If possible, then find $S \circ T$ or $T \circ S$.
12. Let the linear transformation $T: P_3([0, 1]) \rightarrow P_2([0, 1])$ be defined by $(Tp)(x) = p''(x) + p'(x)$. Find the matrix representation of T with respect to the basis $\{1, x, x^2, x^3\}$ and $\{1, x, x^2\}$ of $P_3([0, 1])$ and $P_2([0, 1])$ respectively.
13. Prove that the map $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(x, y, z) = (x + z, x - z, y)$ is invertible.
14. Check whether the following set is orthogonal or not. If not, use the Gram-Schmidt process to find the corresponding orthogonal set.
 - (a). $\{(1, 1), (-1, 1)\}$
 - (b). $\{(0, 10, -10), (1, 2, 2), (2, 0, -1)\}$
 - (c). $\{x^2 + x, x^3, -x^2 + x\}$
 - (d). $\{(1, 2, 4), (3, 2, -2), (2, -3, 1)\}$
16. Define orthonormal set in $\mathbb{R}^3$. Check for the orthonormality of the following sets. If not orthonormal, then find the corresponding orthonormal set using Gram-Schmidt process.
 - (a). $\{(1, 1), (-1, 1)\}$
 - (b). $\{(0, 1, 1), (0, 1, -1), (2, 0, 0)\}$
 - (c). $\{(1, 2, 1), (1, -2, 3), (0, 1, 3)\}$